

Problem Set 31

Evaluating Integrals

1

Evaluate the following definite and indefinite integrals; please show your work. It will sometimes be helpful to simplify your functions prior to looking for an antiderivative.

Although FTOC 2 is very useful for evaluating definite integrals, we have learned other ways of evaluating definite integrals; those methods are still useful for some of the integrals below! Remember that you can always check your indefinite integrals by differentiating them.

(a) $\int \left(\frac{1}{2x} + \frac{e}{\pi x^2} \right) dx$

(d) $\int \frac{u^2 + \ln \pi}{\sqrt[4]{u}} du$

(b) $\int_{-\pi/6}^{\pi/4} \left(\sec(\theta) \tan(\theta) + \frac{\sec^2(\theta)}{\sqrt{3}} \right) d\theta$

(e) $\int_{-0.2}^{0.2} \arcsin x \, dx$

(c) $\int_1^5 |2t - 7| \, dt$

(f) $\int_0^{-2} \left(\sqrt{4 - x^2} + 3e^x \right) dx$

2

(a) Find antiderivatives for the given functions. In other words, for each function f , find a function F such that $F' = f$.

i. $f(x) = e^{3x}$

iii. $f(x) = \left(\sqrt[3]{x^2} + 7x^2 \right) \left(\pi \sqrt{x^5} + e \right)$

ii. $f(x) = \frac{3}{e^x}$

iv. $f(x) = \frac{(3e^x)^2}{\sqrt{5^x}}$

(b) Check your answers to (a) by differentiating them.

3

The Long Trail in Vermont is the oldest long-distance trail in the United States. Suppose we measure positions on the trail relative to Camel's Hump, a peak on the trail. That is, we'll describe positions along the trail by how many miles north of Camel's Hump they are. (So, a location 10 miles south of Camel's Hump would be at position -10 .)

One day, Ali sets out from Camel's Hump at noon for a hike. His velocity t hours after noon is

$$v(t) = \frac{3t^2 + 2t - 56}{24} \text{ miles per hour.}$$

- (a) Find Ali's net change in position between noon and 7 pm. Use a graph to justify your work.
- (b) Find the total distance Ali hiked between noon and 7 pm.
- (c) Describe Ali's hike in words. (We're looking for an answer like, "He walked 1 mile north, then turned around at 1 pm and walked 2 miles south, then ...".)

4

Suppose that a car driver brakes when the car is traveling at 60 mph (which is 88 ft/s). Based on data from the National Highway Traffic Safety Administration, for a typical car, the braking gives the car a constant acceleration of -22 ft/s^2 . How long will it take for such a car to come to a complete stop? How far will the car travel before stopping?

5

Reflect Back. When evaluating integrals, one of the most important skills is to recognize antiderivatives that you can do easily. Evaluate **five** of the integrals below. (The rest are difficult! So, your job is to figure out which are easy and to do those.)

(a) $\int \frac{1}{u^2} du$

(d) $\int \frac{1}{e^u} du$

(g) $\int \tan u du$

(b) $\int \frac{1}{u^2 - 1} du$

(e) $\int u^{-1} du$

(h) $\int \sin u du$

(c) $\int \frac{1}{u^2 + 1} du$

(f) $\int \ln u du$

(i) $\int \sin(u^2) du$